



MatGen: A Realistic Sparse Matrix Generator

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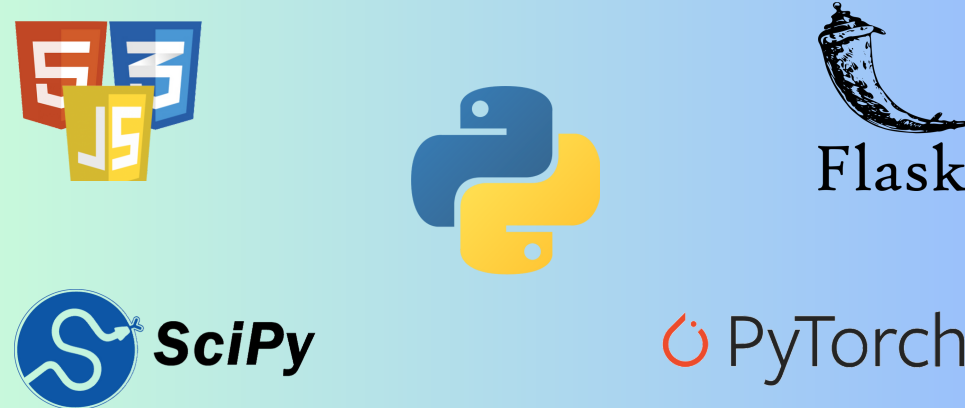
Introduction

MatGen is a Python-based tool that generates custom-made matrix datasets. It helps researchers and developers test their data science models by creating realistic, adjustable data.

Motivation

- Sparse matrices are common in many fields, including medical imaging, structural engineering, recommender systems, natural language processing, computer vision, wireless sensor networks, and astronomy.
- Challenges in gathering complete, high-quality sparse data include high costs, privacy issues, equipment limitations, and missing values—hindering the development and testing of data-driven models with real-world datasets.
- To address this, we created MatGen, a tool that generates synthetic, customizable sparse matrix datasets to help researchers test algorithms under realistic conditions and build more robust solutions without relying solely on scarce real data.

Tech Stack



Results

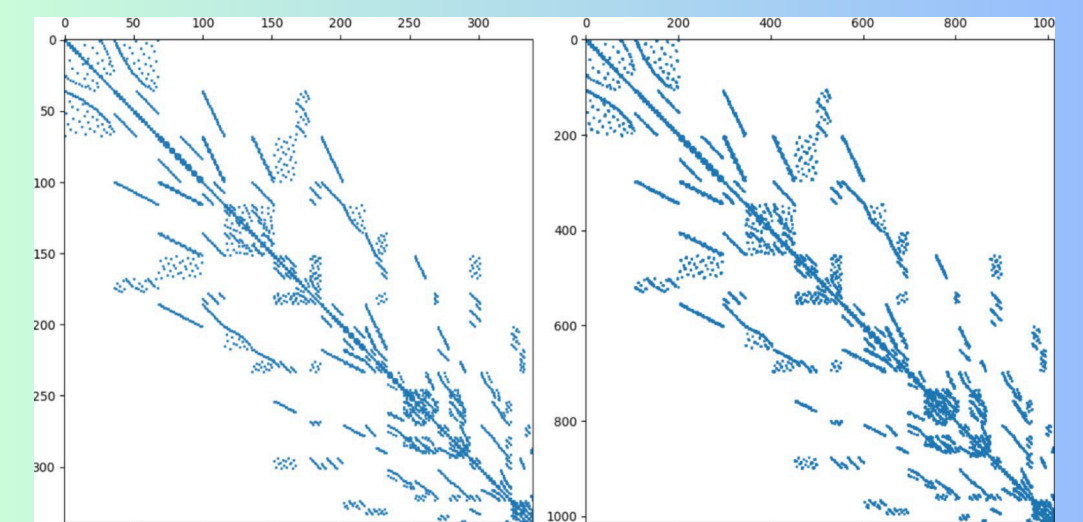


Figure 2: Example visualization of an input matrix and generated matrix

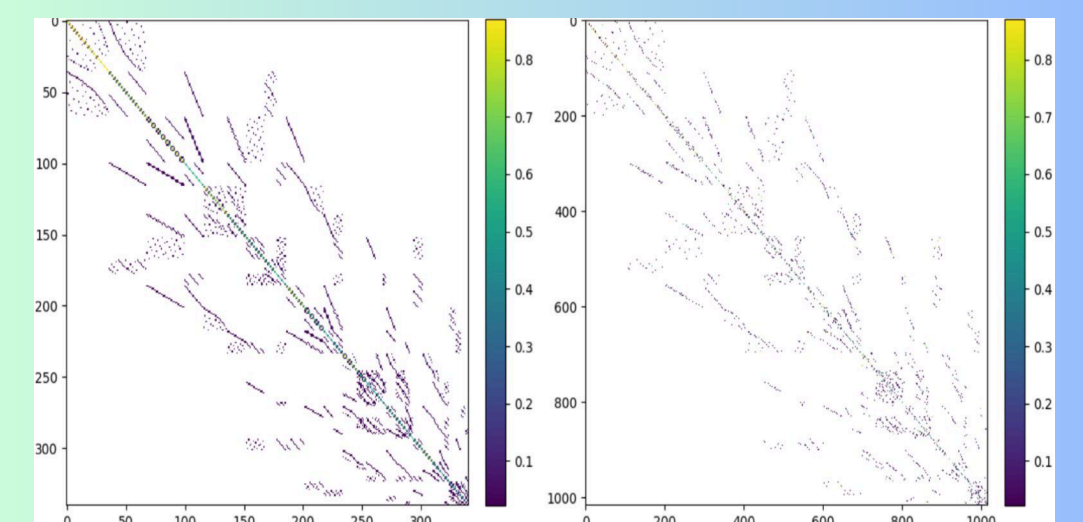


Figure 3: Example heatmaps of an input matrix and generated matrix

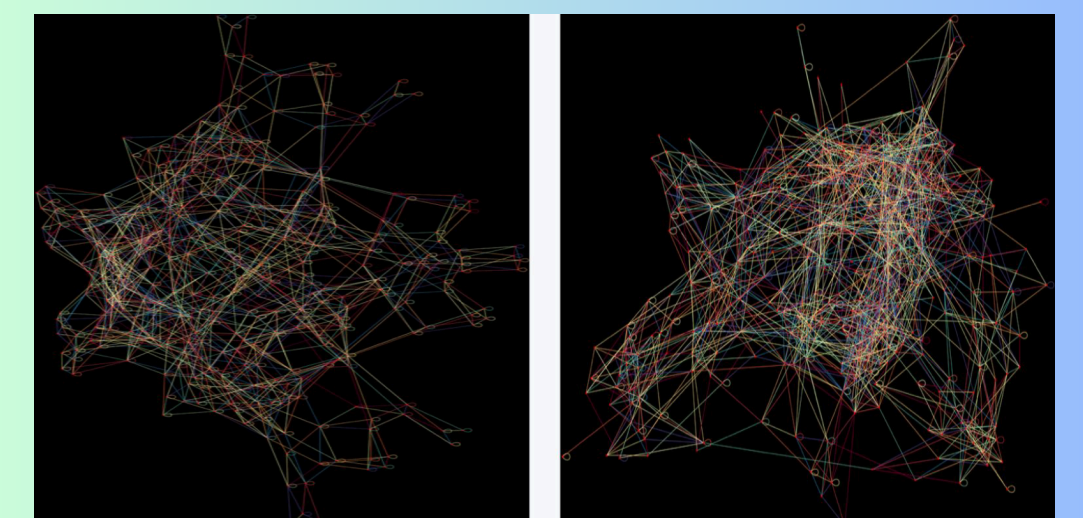


Figure 4: Example graphs of an input matrix and generated matrix

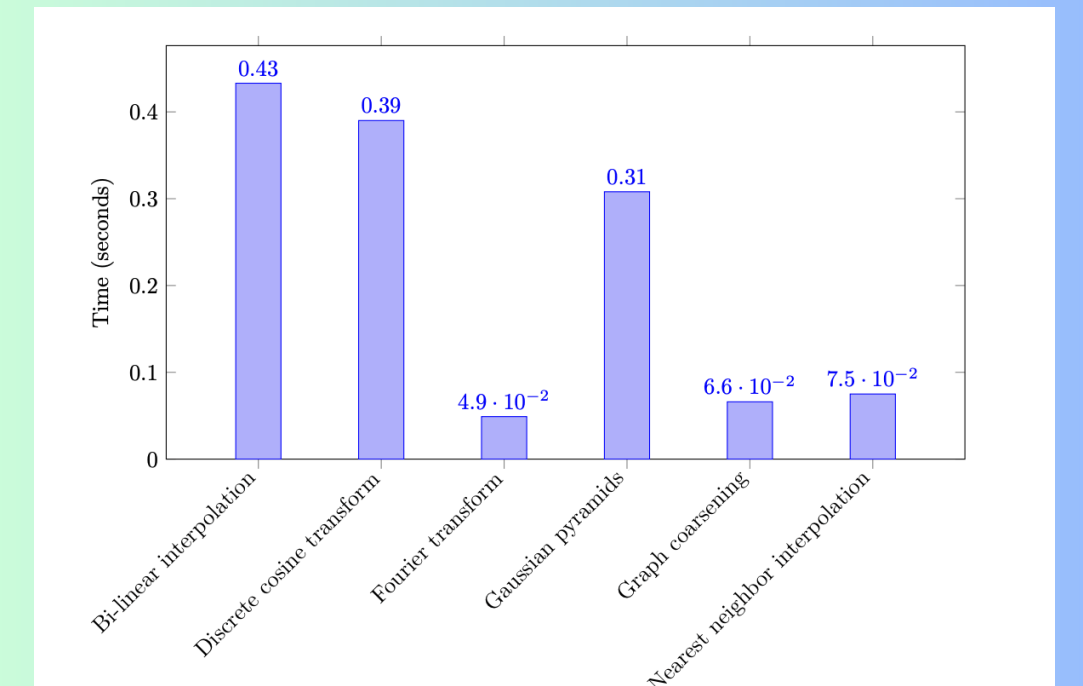


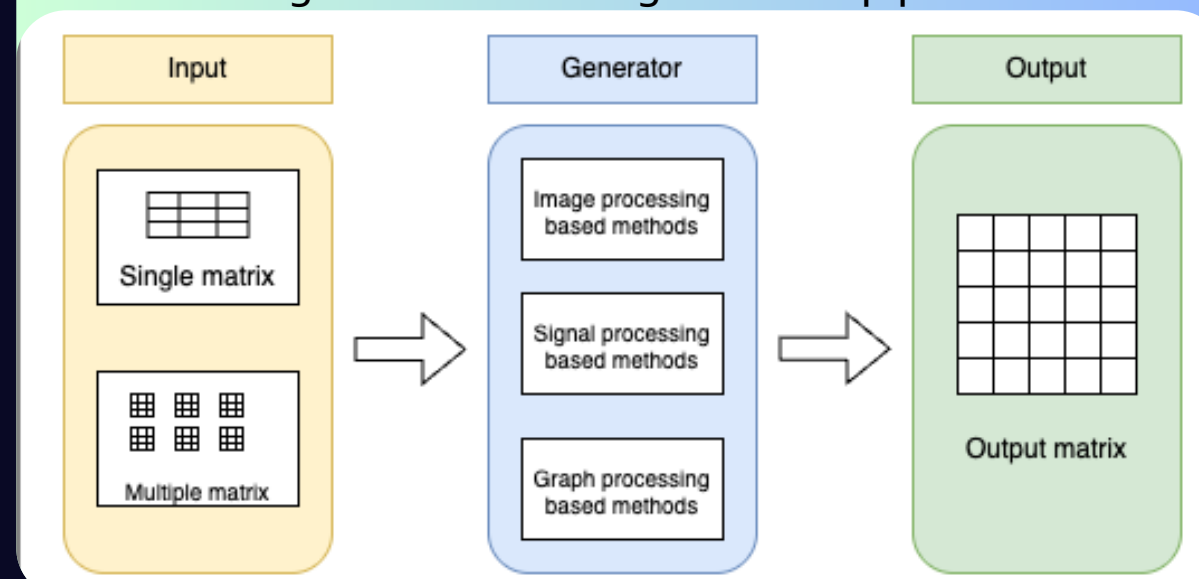
Figure 5: Execution times of different generation methods generating the same matrix

Methodology

We implemented diverse methods from various domains.

- Image Processing:** Bi-linear interpolation, nearest neighbor interpolation, Gaussian pyramids, and Lanczos resampling
- Signal Processing:** Discrete Cosine Transform, Fourier transforms, and Wavelet transforms
- Graph Processing:** Graph coarsening, and Kronecker product

Diagram of the data generation pipeline



Data Collection

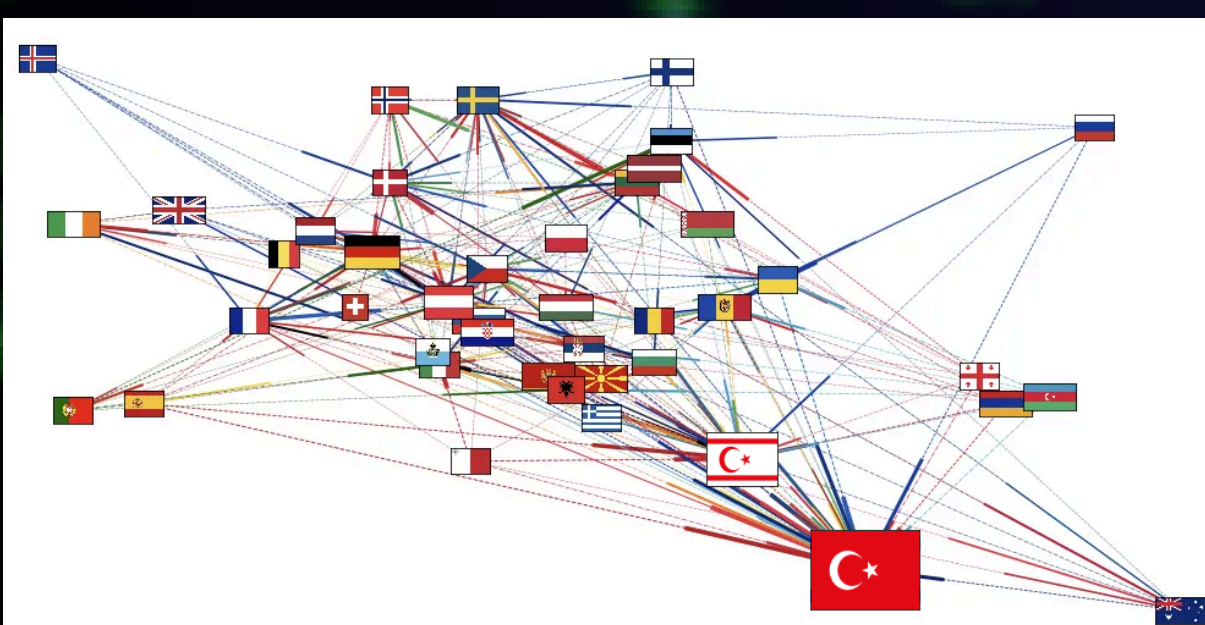


Figure 1: Example graph network (Eurovision Song Contest voting patterns)

